Health data

2026-03-16

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This website includes documentation for lectures and exercises concerning the health data part of the course STA220 Health data and questionnaires. General information about the course is found on the corresponding Canvas page.

Plan

> **Note**
>
> - See Canvas/TimeEdit for schedule.
> - Note that this plan only considers the health data part of the course!
> - See details below regarding the literature
> - Some literature is only used partially (as described in the handouts)
> - The literature is associated with each lecture/session but is not required reading before the session (unless otherwise stated).

> **Warning**
>
> - The plan, as well as the content of each session, is preliminary and may change before the respective session is scheduled.
> - Modifications to lecture slides (and thus indirectly to the lecture handouts) may be made even after the session has taken place. Such changes are only intended to clarify points discussed during the session or to correct mistakes.

Weekday	Time	Shared[1]	Compulsory	Activity	Session	title	subtitle	literature
w4								
Mon	10:15-12	X		Lecture	EL1	Intro	Overwiev of this part of the course	[1, ch. 1], [2], [3]
Wed	08:15-12	X		Lecture	EL2	European legislation	GDPR, EHDS and DA	[4], [1, ch. 4]
w5								
Mon	10:15-12	X		Lecture	EL3	Swedish legislation	Laws and regulation	[5], [6]
w9								
Mon	10:15-12			Lecture	EL4	Tooling	Positron and version control	[7], [8, ch. 4]
	13:15-16			Computer training	ECS1	IDE and version control	Positron, git and GitHub	
Wed	10:15-12			Lecture	EL5	Reproducibility	{targets} and safe environments	[1, ch. 2], [9], [10], [11], [12]
	13:15-16		Compulsory	Seminar	ES1	Ethics and Legality		
w10								
Wed	10:15-12			Lecture	EL6	Data formats	data.table, SQL, DuckDB, parquett, SAS, JSON, API	[1, ch. 2], [13, ch. 21], [14], [15]
	13:15-16		Compulsory	Computer training	ECS2	Pipeline and reproducibility	targets and understanding the data	
w11								

[1] Shared sessin for the whole course (AGE and EB).

Weekday	Time	Shared[1]	Compulsory	Activity	Session	title	subtitle	literature
Mon	10:15-12			Lecture	EL7	Medical coding	ICD, ATC, KVÅ, regex	[1, ch. 3], [13, ch. 15], [16], [17], [18]
	13:15-16		Compulsory	Computer training	ECS3	Data formats and regex		
Wed	11:15-13			Lecture	EL8	Health care registers	From cradle to grave	[19], [20]
	14:15-17			Computer training	ECS4	Data project		
w12								
Mon	10:15-12			Lecture	EL9	Documentation	Quarto	[13, ch. 28-29]
	13:15-16			Computer training	ECS5	Presentation		
Wed	10:15-12			Lecture	EL10	Biggish data		
	13:15-16			Computer training	ECS6			
w13								
Mon	08:15-12	X		Lecture	EL11	Repetition	Repetition before DISA exam	
	13:15-16		Compulsory	Seminar	ES2	Project presentations		

[1] Shared sessin for the whole course (AGE and EB).

Software

To complete the exercises and assignments for this course you need to install the following software (which are free and available for all major operating systems):

- R
- Positron

- Git

Some additional recommended software (also free but not mandatory for the course):

- GitHub Desktop
- GitHub CLI

Positron extensions

Positron is an integrated development environment (IDE) for R. It makes it easy to write and execute R code, manage projects, and visualize data. It is made by Posit (formerly RStudio) PBC and is free for individual use.

Positron is based on Code OS, an open source version of VS Code from Microsoft. As such, it supports a wide range of extensions and customization options.

We will use GitHub Pull Requests. Search for it in the Extensions pane in Positron and install it.

Instructions for Git in Positron

Litterature

Course books are available for GU students through the "O'Reilly Learning for Higher Education". Instructions for access

Primary course book. Not every chapter of the book will be discussed during the course, and some programming examples in the book are written in languages that we will not use, as our programming focus is mainly on R.

Recommended books.

Additional mandatory reading

PDF versions of scientific articles are found in Canvas.

DISA exam

- The literature listed in the table above (literature column) is used for examination.
- Some references, however, are only partially used (see handouts and lecture slides).
- Some of the literature is examined implicitly as part of the project part of the course.

The DISA exam will examine:

- The course book A. Nguyen [1] (selected parts of chapter 1-4)
- All PDF:s found in the Canvas module
  - Most tables and detailed methods sections can be skipped
- Lecture handouts, excluding:
  - EL3 (examined only as part of ES1)
  - EL8 where A. Hiyoshi [19] (one of the PDF:s) is a sufficient reference
  - EL10 which is formally not examined but which might be useful for the project component of the course.

The written examination will assess your understanding of the main themes covered in the course. The purpose of the exam is not to test memorization of details, but your ability to explain concepts and reflect on methodological issues in health data research.

Structure of the exam

Section A – Conceptual understanding

In this section you will answer a number of short questions about key concepts from the course. Your answers should be brief (usually **1–3 sentences**). The goal is to demonstrate that you understand the terminology and the purpose of the concepts discussed in the course.

Section B – Applied understanding

In this section you will be asked to explain or interpret practical situations related to health data analysis. Your answers should be somewhat longer (typically **4–6 sentences**) and demonstrate that you understand how the concepts from the course apply in real research settings. The goal here is to show that you can **connect course concepts to practical research situations**.

Section C – Reflection and synthesis

The final question asks you to reflect on the broader research workflow in modern health data analysis. This part of the exam focuses on your ability to combine ideas from several parts of the course. A good answer should show that you can reason about how these elements interact in real research projects.

Podcast episode

This "podcast episode" has been generated by NotebookML based on the lecture handouts and course literature. It might provide a helpful summary of the course.

Bibliography

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